

Hypomagnesaemia in adults

Derbyshire Shared Care Pathology Guidelines on hypomagnesaemia, hypocalcaemia and hypokalaemia January 2024, BNF January 2024, Int J Nephrol Renovasc Dis 2014;7:219, Clin Biochem Rev 2003;24:47, Nutrients 2015;7:8199, Pharmacol Res Perspect 2021;9:e00829, J Electrocardiol 2023;80:17



Red Whale

GEMS
Guidelines & Evidence Made Simple

**If hypomagnesaemia is present, calcium and/or potassium may also be low.
If this is the case, replacing the magnesium can help normalise the calcium/potassium.**

Do not use this GEMS if CKD 4 or 5: seek renal advice.

Magnesium ≤ 0.4 mmol/l

Likely medical emergency.
Discuss with medics that day:
IV treatment required.

**Admit.
IV replacement
needed.**

Magnesium >0.4 to <0.7 mmol/l

Any symptoms of hypomagnesaemia?

Symptoms similar to hypocalcaemia. Having ANY symptom is significant:

Muscle cramps, aches or pains
Neurological symptoms: paresthesia, tetany or seizures
Depression, neuropsychiatric symptoms
Cardiovascular symptoms: arrhythmia, heart failure

No

Do same-day ECG if possible

Hypomagnesaemia can cause prolonged PR interval, prolonged QT interval, wide QRS, T-wave inversion, appearance of U waves, supraventricular or ventricular arrhythmia or torsade de pointes
(QT prolongation and ventricular ectopics can precede torsade de pointes).
Hypokalaemia can co-exist with low magnesium. Hypokalaemia can cause ECG abnormalities which also require admission. See the hypokalaemia GEMS for more info.

ECG
abnormal

ECG normal

**If magnesium >0.4 to <0.7 mmol/l AND asymptomatic AND ECG normal (if available)
OR
secondary care advised primary care management,
give oral replacement in primary care.**

Magnesium aspartate 10mmol sachets: 1–2 sachets/day (avoid if eGFR <30).

Some magnesium glycerophosphate chewable tablets are also licensed (avoid if creatinine clearance <30).
Derbyshire guideline recommends co-magaldrox 10–20ml orally four times a day (unlicensed combination of magnesium and aluminium) for up to 6–8w, checking magnesium every 1–2w during replacement.
Commonest side-effect of magnesium replacement is diarrhoea; if it occurs, reduce the dose.

Assess for a cause

Most commonly due to GI or renal causes:

Renal causes:

- Drugs: PPIs (most commonly), diuretics (especially loop diuretics), cytotoxic drugs, aminoglycosides, immunosuppressants, theophylline.
- Osmotic diuresis (diabetes).
- Renal tubular acidosis.
- Inherited disorders, e.g. Gitelman's syndrome (rare!).

Other causes:

- Alcohol abuse.

GI causes:

- Reduced absorption: malabsorptive conditions (e.g. coeliac disease), chronic diarrhoea, laxative misuse, fistula, short bowel syndrome, pancreatitis.
- Reduced intake: dietary deficiency.

Endocrine causes:

- Hyperthyroidism, parathyroid disease, hyperaldosteronism.

Review medications: stop PPI and review diuretics.

If causative agent advised/prescribed by secondary care, liaise with specialist team (e.g. oncology).

If unknown cause, refer to/seek advice and guidance from endocrinology.

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